

PREFINAL

Course: SADP (55)

Set: B | Time: 3 Hours | Max Marks: 100 | Date: 16-03-2026

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

GENERAL INSTRUCTIONS:
ANSWER ALL THE QUESTIONS

SECTION A: MCQs

1. Which design pattern is used in Lexi to represent the document's physical hierarchy, allowing both individual characters and complex compositions like rows to be treated uniformly? **[6 Marks]**

- a) Strategy Pattern
- b) Bridge Pattern
- c) Composite Pattern
- d) Command Pattern

2. In Lexi's formatting system, which class acts as the root of the hierarchy that encapsulates the line-breaking algorithm? **[6 Marks]**

- a) Glyph
- b) Composition
- c) WindowImp
- d) Compositor

3. How does Lexi add visual elements like borders and scroll bars to its user interface without creating an excessive number of subclasses? **[6 Marks]**

- a) By using Transparent Enclosure and the Decorator Pattern
- b) By using the Abstract Factory Pattern to create widgets
- c) By implementing a separate WindowImp hierarchy
- d) By using the Command Pattern to encapsulate UI requests

4. What is the purpose of the GUIFactory abstract class in Lexi's design? **[6 Marks]**

- a) To manage the history list for undo and redo operations
- b) To define an interface for creating platform-specific UI widgets
- c) To encapsulate the low-level window system calls

d) To define the recursive composition of characters and rows

5. Which pattern decouples Lexi's windowing abstraction from its platform-specific implementation, allowing them to vary independently? **[6 Marks]**

- a) Iterator Pattern
- b) Strategy Pattern
- c) Bridge Pattern
- d) Visitor Pattern

SECTION B: SHORT ANSWERS

6. Describe how the Glyph abstract class facilitates the recursive composition of a Lexi document. **[6 Marks]**

7. Explain the role of the Command pattern in Lexi's user operations and how it supports undo functionality. **[6 Marks]**

8. How does the combination of the Iterator and Visitor patterns allow Lexi to perform spelling checks without modifying Glyph classes? **[6 Marks]**

9. Distinguish between the roles of the Window and WindowImp hierarchies in Lexi's windowing system support. **[6 Marks]**

10. What specific problem does the Strategy pattern solve in Lexi's formatting design, and what classes are involved? **[6 Marks]**

SECTION C: LONG ANSWERS

11. Analyze how the Composite pattern enables Lexi to treat text and graphics uniformly and the benefits this uniformity provides to the system. **[8 Marks]**

12. Evaluate the implementation of the Strategy pattern in Lexi's formatting system regarding the "open-closed" principle. **[8 Marks]**

13. Describe the "Transparent Enclosure" concept used in Lexi's UI embellishment and explain how it prevents a "combinatorial explosion of subclasses." **[8 Marks]**

14. Explain how Lexi uses the Abstract Factory pattern to remain portable across different Look-and-Feel standards like Motif and PM. **[8 Marks]**

15. Discuss how Lexi's design for User Operations handles the challenges of scattered functionality and the requirement

for robust command history management.

[8 Marks]

***** END OF QUESTION PAPER *****